

Where, Not What

The determinants of city size in the Holy Roman Empire and Central Europe, 1200–1500

A quantitative study of 415 imperial cities (2,262 across Europe), the Viabundus medieval transport network, and the dated commercial privileges of the *Deutsches Städtebuch*.

Abstract. Why did some medieval cities swell into metropolises while their neighbours stagnated? Drawing on population estimates for 2,262 European cities, the reconstructed medieval road-and-water network (Viabundus), and dated grants of staple rights, fairs, town charters and market rights for 2,390 imperial cities, we test the commonest explanation — that commercial privileges and institutions made cities great — and find it does not survive measurement. The privileges historians celebrate were awarded to towns that had *already* risen; once we compare a chartered town to comparable towns over the same years, the growth effect of a staple right, a fair, a charter, or a market right is statistically indistinguishable from zero. What actually fixed the urban hierarchy was set long before: roughly seven-tenths of a city's size in 1500 was determined by its size centuries earlier, and a city's population in **800 CE still predicts its size 700 years later**. The one force that reliably moved a town up or down was **access to navigable water** — worth about **+15% growth per century** and decisive in the reshuffling that followed the Black Death. The system is captured by a single equation of near-random growth anchored on geography. Cities grew great because of **where** they sat and **how early** they began — not because of what their rulers granted them.

1. The puzzle, and why it resists a simple formula

Economic history has long sought the "recipe" for urban success: the right charter, the right market, the right institutions. If such a recipe existed, cities with more of the right ingredients should have grown faster. We test this directly and reach the opposite conclusion — and, importantly, we can show *why* a deterministic recipe cannot exist.

The first fact any theory must confront is that, from one century to the next, **a city's growth is almost unrelated to anything about the city** — including its own size. Figure 1 plots each city's growth from 1200 to 1500 against its starting size. The cloud is essentially flat.



Figure 1. Growth 1200→1500 against starting size. The near-zero slope means big and small cities grew at essentially the same average rate — statisticians call this **Gibrat's law**. A city's size explains only about 2% of its growth.

Plain-language note — what " R^2 " means. Throughout, R^2 is the share of the variation a model explains, from 0 (nothing) to 1 (everything). An R^2 of 0.02 means the factor accounts for 2% of what happened and 98% is left unexplained. When century-to-century growth has an R^2 near 0.02 against every candidate factor, there is simply very little systematic signal for *any* equation to capture. This is not a defect of the data; it is a property of how cities grow.

This is the deep reason a factor-based growth equation underperforms: most of what happens to a city in a century is idiosyncratic. The productive question is therefore not "what predicts growth?" but "what fixes the *hierarchy* — the relative ranking of cities — and did the institutions we credit actually cause it?"

2. Data

Two independent population reconstructions

Medieval city populations are not measured but *reconstructed* — pieced together from tax rolls, hearth counts, militia lists, church records and chronicles, then converted to inhabitants with demographic multipliers. Because the exercise is inevitably uncertain, we run our entire analysis on the **two most widely used reconstructions of European city populations**, built by different scholars from different sources, and report whether the conclusions survive the choice of source.

- **Bairoch (1988)** — Paul Bairoch, Jean Batou & Pierre Chèvre, *La population des villes européennes de 800 à 1850*. The foundational compilation of the field: roughly 2,200 European cities at centennial snapshots from 800 to 1850, synthesised from a vast body of local and secondary historical sources. For a generation it has been *the* standard reference for quantitative work on European urbanisation.
- **Buringh (2021)** — Eltjo Buringh, *The Population of European Cities from 700 to 2000* (Research Data Journal for the Humanities and Social Sciences). A later, georeferenced reconstruction of 2,262 cities that revises and extends Bairoch and Chandler with newer local sources, and — crucially for us — attaches to each city its **coordinates, elevation**, and a **first-nature transport classification** (river / coastal / landlocked, with sea-catchment). It underlies the influential study of Bosker, Buringh & van Zanden (2013).

The two agree closely — the correlation of their (log) city sizes in 1500 is **0.96** (Figure 8) — but they were assembled independently, so agreement of our *results* across them is a meaningful robustness test. We take Buringh as the primary panel (it carries the geography we need) and Bairoch as the check, and study cities of $\geq 1,000$ inhabitants throughout.

The transport network and the privileges

- **The transport network:** Viabundus — a georeferenced reconstruction of the late-medieval road, river and sea network of Northern Europe, including **dated** staple-right, fair and toll nodes.
- **Commercial privileges:** Cantoni, Mohr & Weigand (2020), *Princes and Townspeople* — dated town charters (*Stadtrecht*) and market rights (*Marktrecht*) for all 2,390 cities of the *Deutsches Städtebuch*, with legal families (Magdeburg law, Lübeck law, ...).

3. The central result: privileges were badges of success, not its cause

Consider staple rights (*Stapelrecht*) — the prized privilege forcing merchants passing through a town to unload and offer their goods for sale. Towns with a staple right were, on average, **53% larger** than towns without one (the red bars in Figure 2). Taken at face value, this looks like proof that the privilege built the city. It is not.

The problem is **timing**. A staple right was granted *to* a town that had already become an important junction. To separate cause from consequence we ask a sharper question: when a town acquired a privilege, did its growth *accelerate* relative to comparable towns that did not yet have it? This is a **difference-in-differences** comparison.

Plain-language note — difference-in-differences. Imagine two similar towns. One receives a staple right in 1300, the other does not. If the privilege causes growth, the first town should pull ahead *after* 1300 by more than it was already pulling ahead *before* 1300. We measure the "extra" acceleration and net out whatever both towns were doing anyway. If the granted town grows no faster after the grant than before — and no faster than its untreated peers — the privilege did not cause the growth.

Across **four** distinct privileges from **two** independent datasets, the causal (blue) estimate collapses to zero, and every confidence interval spans zero:

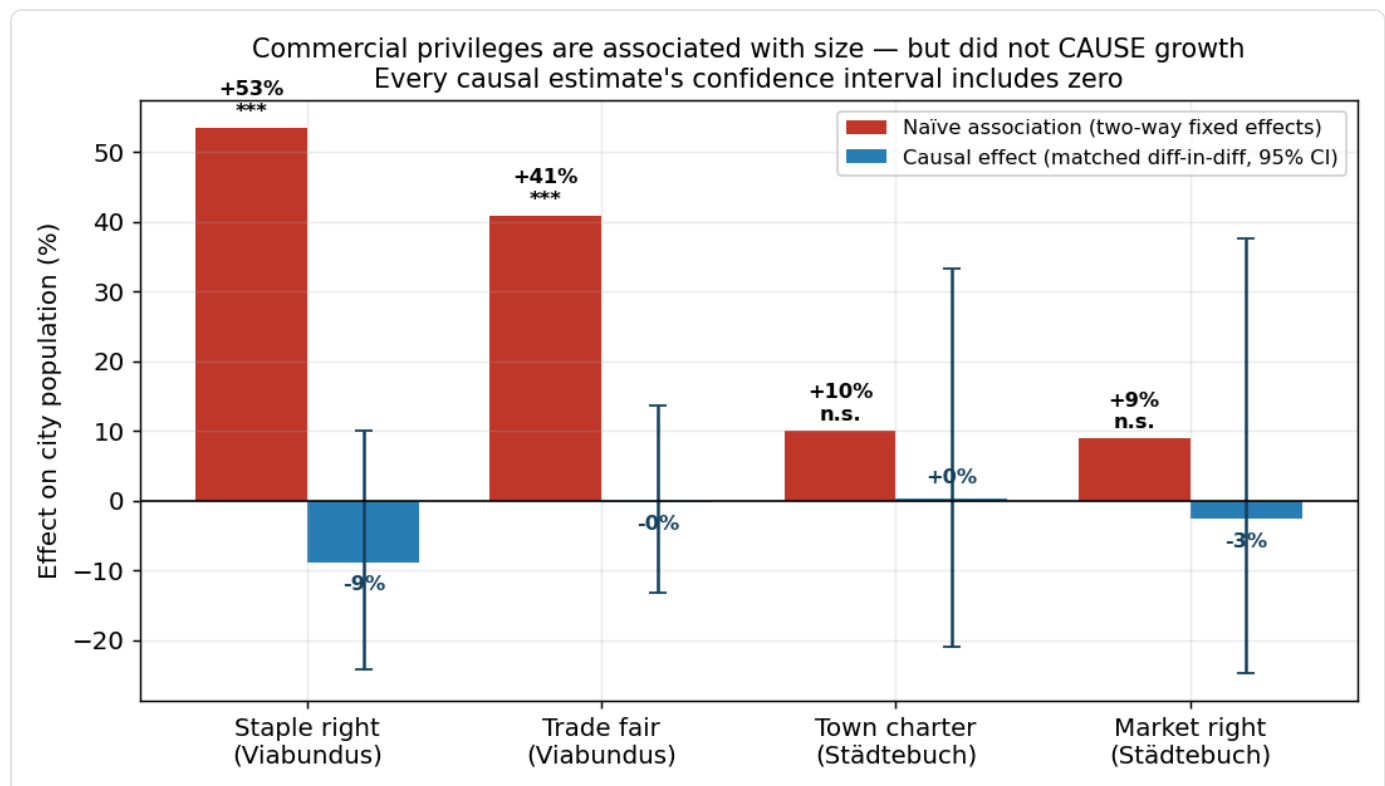


Figure 2. Red: the raw size gap between towns with and without each privilege (two-way fixed-effects association). Blue: the causal effect once we compare like with like over the same years (matched difference-in-differences, with 95% confidence bars). The +53% staple "effect" and +41% fair "effect" vanish; charters and markets were never even statistically significant.

Privilege	Source	Raw size gap	Causal effect	95% CI	Growth before grant	Growth after grant
Staple right (Stapelrecht)	Viabundus	+53% ***	-9%	[-24%, +10%]	+0.31	+0.34
Trade fair (Messe)	Viabundus	+41% ***	-0%	[-13%, +14%]	+0.29	+0.28
Town charter (Stadtrecht)	Städtebuch	+10% n.s.	+0%	[-21%, +33%]	+0.24	+0.21
Market right (Marktrecht)	Städtebuch	+9% n.s.	-3%	[-25%, +38%]	+0.26	+0.36

Table 1. The two right-hand columns are the giveaway: growth in the century *before* a privilege was granted is essentially identical to growth in the century *after*. Nothing changed at the moment of the grant. (***) = significant at 1%; n.s. = not significant.)

Finding 1. Staple rights, fairs, town charters and market rights show no causal effect on city growth. The strong raw associations are entirely *selection*: privileges were conferred on towns that had already grown. This mirrors the well-known result (Bosker, Buringh & van Zanden 2013) that the "bishopric" advantage disappears once city fixed effects absorb the fact that bishops were placed in already-important cities.

Cologne against Hamburg: the pattern in two cities

The *Städtebuch* record makes the logic vivid. **Cologne**, the empire's largest city (40,000 in 1200), was a Roman *colonia* and never received a medieval town charter — it was a city before charters existed. **Hamburg**, a small place of 2,000 in 1200, received its charter in 1215 and a staple right, and grew fifteen-fold. Yet **Mainz** held both a charter (1300) and a staple right and *shrank* from 9,000 to 6,000. The privileges neither made Cologne nor saved Mainz; they tracked, imperfectly, an underlying geography of trade (Table 3).

4. What actually fixed the hierarchy: deep history and geography

If institutions are not the cause, what is? We decompose the variation in city size in 1500 into three sources: a city's **inherited size** (its size in 1200, i.e. path dependence), its **deep origin** (its size already in 800 CE), and its **first-nature geography** (river/coast access, sea basin, elevation). A Shapley decomposition — a fair way of splitting shared credit among overlapping causes — gives:

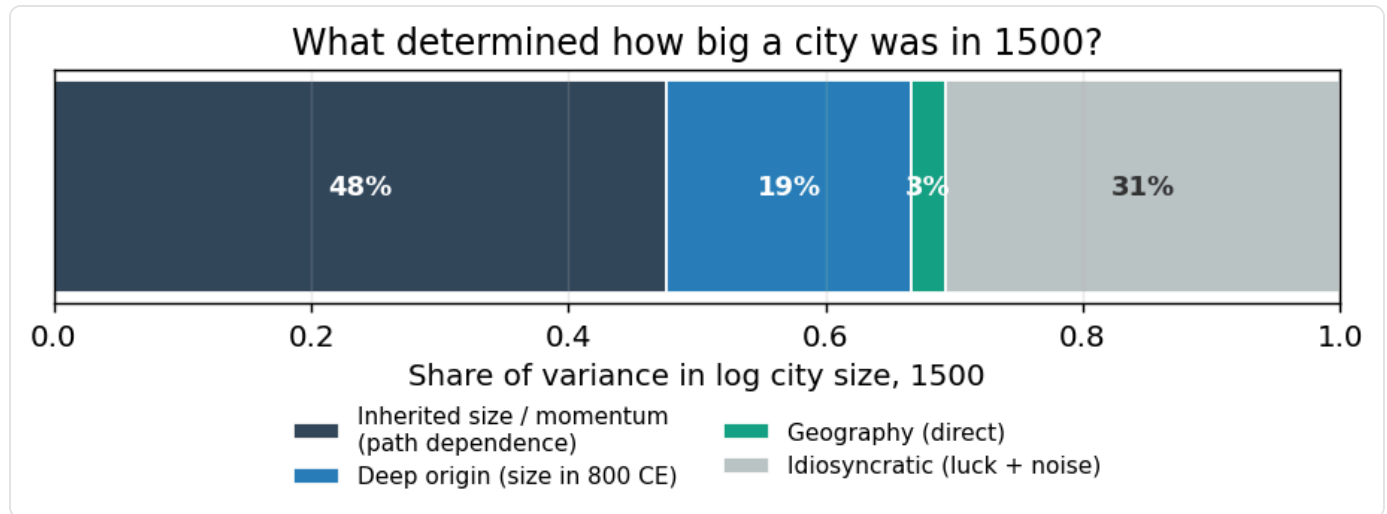


Figure 3. What determined a city's size in 1500. Inherited size and deep origin — i.e. *history* — dominate. Geography's *direct* share is small only because its work was already done: geography set the early sizes, and persistence carried them forward. Nearly a third of the outcome is irreducibly idiosyncratic.

The persistence is astonishing in its reach. A city's size in **800 CE** — the age of Charlemagne — still explains **40%** of its size in 1500 (Figure 4). The urban map of the Holy Roman Empire in 1500 was, to a first approximation, drawn seven centuries earlier.

The urban hierarchy was locked in centuries early
A city's size in 800 CE still explains 40% of its size 700 years later

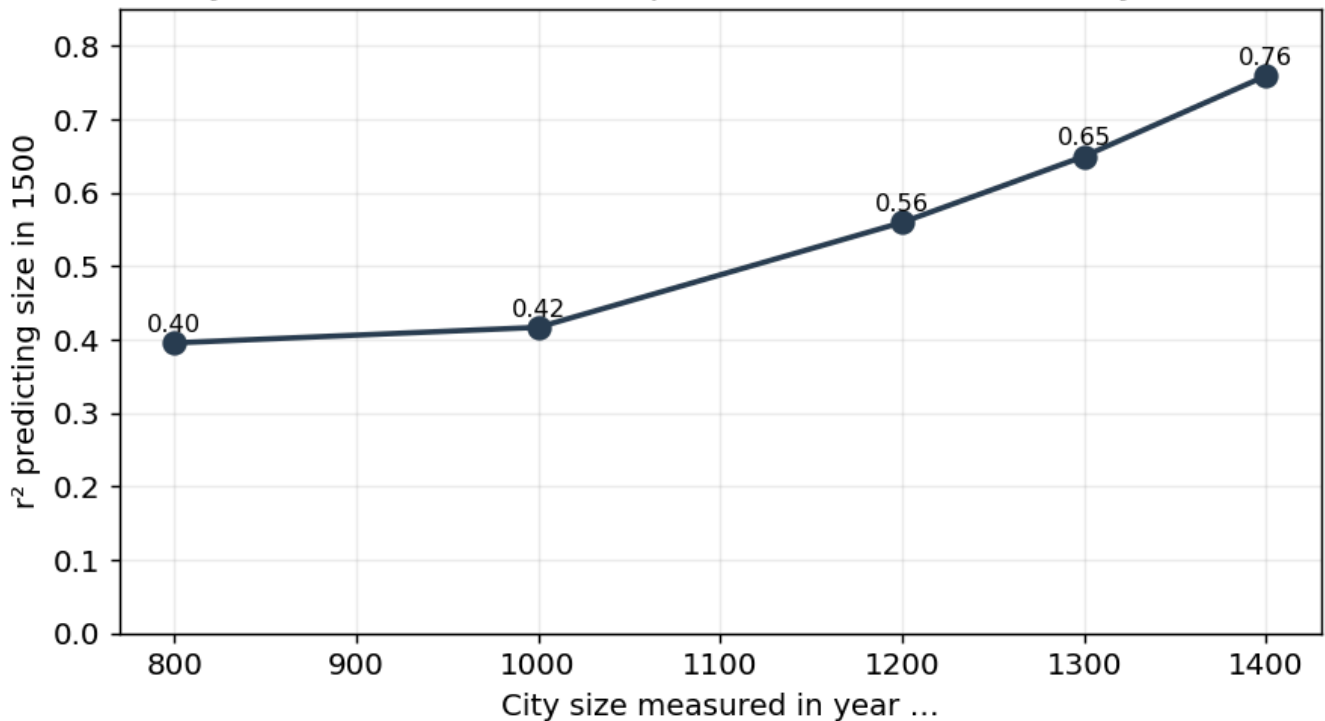


Figure 4. How strongly a city's size in each past year predicts its size in 1500. Even the Carolingian-era ranking of 800 CE survives into the Renaissance.

Finding 2. About 70% of a city's 1500 size was already locked in by inherited size and deep origin. The medieval urban hierarchy was overwhelmingly **path-dependent**: history, not policy, was destiny.

5. The one lever that moved cities: water

Geography acts through one channel above all: **access to navigable water**. A city on a river or coast grew about **15% more per century** than a landlocked one (95% confidence interval +7% to +23%). Because a town cannot relocate to the sea, this relationship is plausibly causal in a way the endogenous privileges are not.

But water's power was not constant — it was switched on by a change in the trade regime. Resolving the coastal premium by sea basin and century (Figure 5) reveals a sharp turn. During the Black Death (1300–1400), coastal cities — especially Mediterranean ports, the plague's entry points — did *worse*. After 1400, every sea basin turned strongly positive, and the Atlantic and

North Sea rose fastest: the beginning of the maritime economy that would define the following centuries.

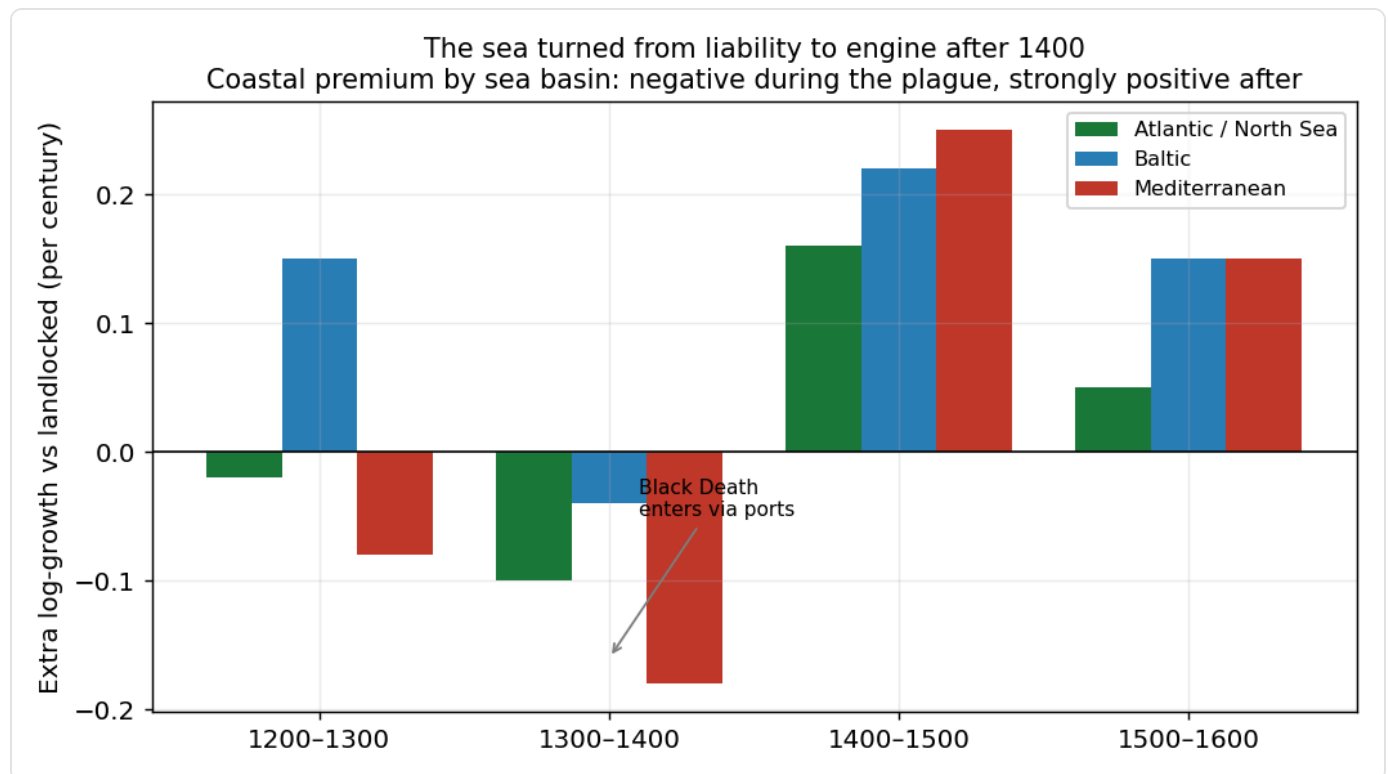


Figure 5. The extra growth of coastal cities (vs landlocked), by sea basin and century. The sea was a liability during the plague and an engine after it.

The consequences are written directly onto the map of winners and losers. Figure 6 ranks the cities that most exceeded, or fell short of, the size their 1200 population alone would predict. The over-performers are almost entirely Atlantic and North-Sea ports — Hamburg, Amsterdam, Antwerp, Danzig, Utrecht; the under-performers are Mediterranean and landlocked river towns.

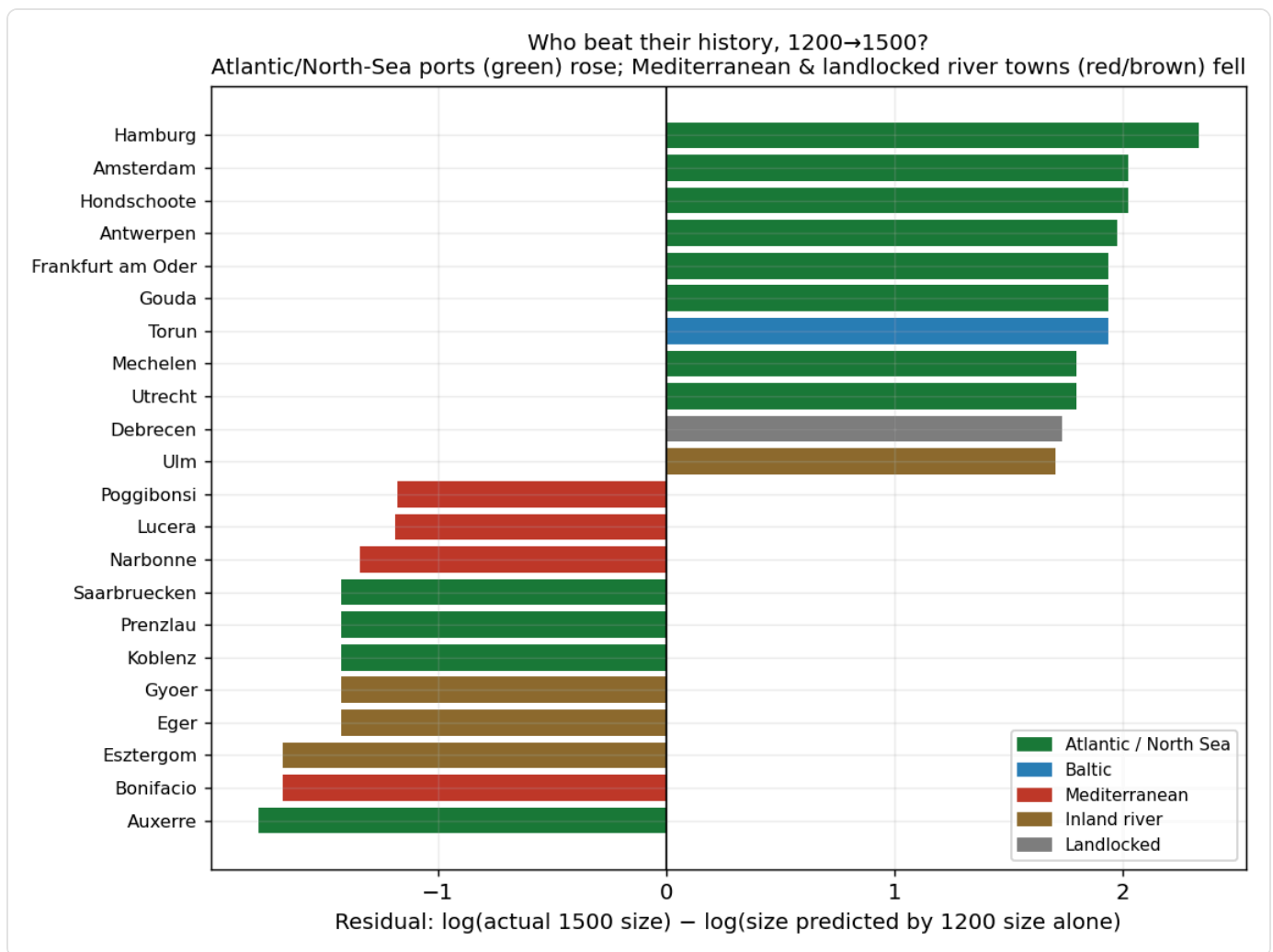


Figure 6. Who beat their own history. Bars show how much bigger (right) or smaller (left) a city was in 1500 than its 1200 size predicts, coloured by sea basin.

Rose fastest above expectation

- **Hamburg** (2,000→30,000; Atlantic/N.Sea)
- **Amsterdam** (1,000→12,000; Atlantic/N.Sea)
- **Hondschoote** (1,000→12,000; Atlantic/N.Sea)
- **Antwerpen** (3,000→30,000; Atlantic/N.Sea)
- **Frankfurt am Oder** (1,000→11,000; Atlantic/N.Sea)
- **Gouda** (1,000→11,000; Atlantic/N.Sea)
- **Torun** (1,000→11,000; Baltic)

Fell furthest below expectation

- **Auxerre** (10,000→2,000; Atlantic/N.Sea)
- **Bonifacio** (4,000→1,000; Mediterranean)
- **Esztergom** (4,000→1,000; Inland river)
- **Eger** (3,000→1,000; Inland river)
- **Gyoer** (3,000→1,000; Inland river)
- **Koblenz** (3,000→1,000; Atlantic/N.Sea)
- **Prenzlau** (3,000→1,000; Atlantic/N.Sea)

- **Mechelen** (3,000→25,000;
Atlantic/N.Sea)

- **Saarbruecken** (3,000→1,000;
Atlantic/N.Sea)

Table 2. The exceptions are instructive: the two landlocked over-performers are Hondschoote (a Flemish textile boom-town) and Debrecen (the hub of the Hungarian cattle trade) — specific export industries, not general institutions.

Twenty-one cities, side by side

City	Pop 1200	Pop 1500	Growth	Water	Charter	Staple	vs. history	
Cologne	40,000	45,000	+13%	✓	—	✓	+0.11	Roman-origin — predates charters
Nuremberg	7,000	38,000	+443%	✓	1200		+1.47	
Augsburg	13,000	30,000	+131%	✓	—		+0.69	Roman-origin — predates charters
Danzig	5,000	30,000	+500%	✓	1224	✓	+1.53	
Hamburg	2,000	30,000	+1400%	✓	1215	✓	+2.33	
Antwerpen	3,000	30,000	+900%	✓	—	✓	+1.98	Low Countries — outside Städtebuch
Utrecht	3,000	25,000	+733%	✓	—	✓	+1.79	Low Countries — outside Städtebuch
Lübeck	11,000	25,000	+127%	✓	1160	✓	+0.66	
Regensburg	16,000	22,000	+37%	✓	1207		+0.20	
Vienna	17,000	20,000	+18%	✓	—		+0.05	Roman-origin — predates charters
Strasbourg	10,000	20,000	+100%	✓	—		+0.52	Roman-origin — predates charters
Erfurt	15,000	19,000	+27%	✓	—	✓	+0.11	no formal medieval charter
Bremen	8,000	18,000	+125%	✓	1186	✓	+0.61	
Magdeburg	16,000	18,000	+13%	✓	1188	✓	-0.00	
Ulm	2,000	16,000	+700%	✓	1274		+1.70	
Amsterdam	1,000	12,000	+1100%	✓	—	✓	+2.02	Low Countries — outside Städtebuch
Frankfurt	4,000	12,000	+200%	✓	1297		+0.81	
Rostock	3,000	10,000	+233%	✓	1218	✓	+0.88	
Basel	5,000	10,000	+100%	✓	—		+0.43	Roman-origin — predates charters
Leipzig	3,000	7,000	+133%	✓	1156	✓	+0.52	
Mainz	9,000	6,000	-33%	✓	1300	✓	-0.60	

Table 3. "vs. history" is the residual from Figure 6 (how much a city beat or missed its 1200-implied size). Note the pattern: the great Roman cities (Cologne, Augsburg, Vienna, Basel, Strasbourg) carried **no medieval charter**; the explosive growers (Hamburg, Danzig, Amsterdam, Antwerp) were water ports; and holding a charter and staple did not save Mainz (−0.60). Every city here is on water — because in this region, being a city of any size in 1200 already required it.

6. One equation for the whole system

All of the above — the near-random growth, the deep persistence, the pull of geography — is captured by a single law of motion for city size. Writing **P** for population and **log** for the natural logarithm (which turns multiplicative growth into additive steps), the size of city *i* in the next century is:

$$\log P_{\text{next}} = a_t + 0.915 \cdot \log P_{\text{now}} + 0.06 \cdot \text{water} + \text{noise}$$

Reading the equation, term by term.

- **0.915 · log P_{now}** — next century's size is almost entirely this century's size. The coefficient **0.915** (just below 1) means cities are extremely "sticky": whatever a city is, it will still be, a century later. This single number produces the deep persistence of Figure 4.
- **a_t** — a shock common to *all* cities in a given century (the Black Death made it strongly negative around 1350; recovery made it strongly positive after 1400). These are era-wide tides, not city-specific.
- **0.06 · water** — the small, steady geographic nudge upward for water-access cities.
- **noise** — a large idiosyncratic term. It is *five times* larger than the systematic pull (technically, the ratio of the reversion force to the noise is 0.21), which is exactly why growth looks random (Figure 1) and why no factor equation can predict it.

When we run this equation forward from the real cities of 1200, it reproduces the observed persistence of the hierarchy and its overall shape. It also reveals what it *cannot* produce: the real hierarchy grew **more top-heavy** over time — the largest cities pulled away (the "Zipf" exponent fell from 1.26 toward 1.0) — whereas pure random growth predicts mild equalisation (Figure 7). That gap is the fingerprint of a genuine **agglomeration** advantage at the very top: beyond a point, size itself attracted more size.

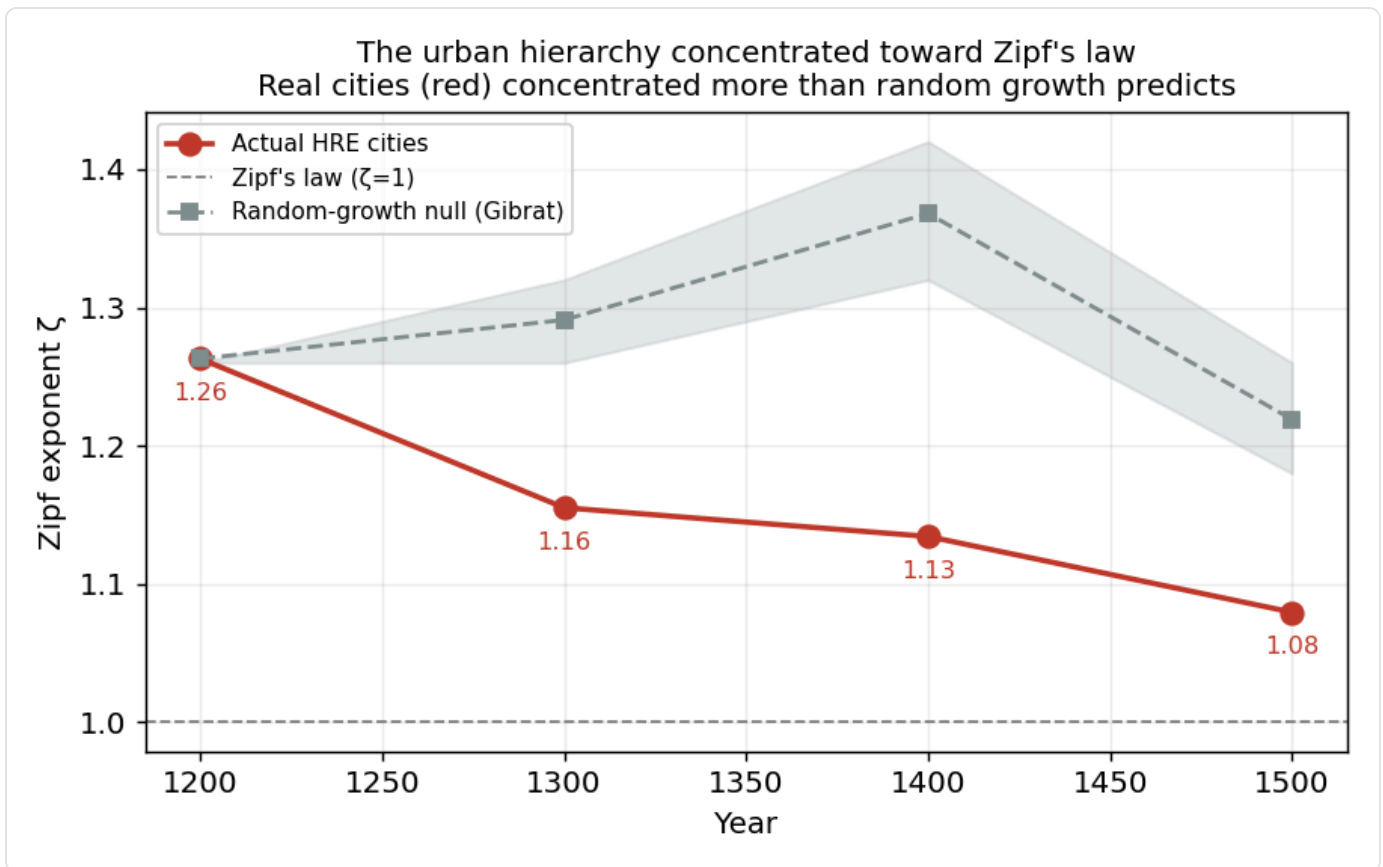


Figure 7. Concentration of the urban hierarchy. A lower value means the largest cities loom larger. Real cities (red) concentrated faster than random growth (grey) predicts — mild but real increasing returns at the top.

Finding 3. Medieval city size follows a near-random-walk anchored on geography, plus era-wide shocks, plus a weak agglomeration pull at the largest cities. It is a *stochastic law*, not a factor formula — which is why the search for a deterministic "recipe" was bound to disappoint.

7. Robustness: the two standard reconstructions tell the same story

Every result above is computed on the Buringh panel. We now recompute the core findings on the independent Bairoch (1988) panel — same cities, same geography, same privileges, only the population figures swapped — to confirm they are not an artefact of one scholar's estimates. First, the two sources substantially agree on the sizes themselves:

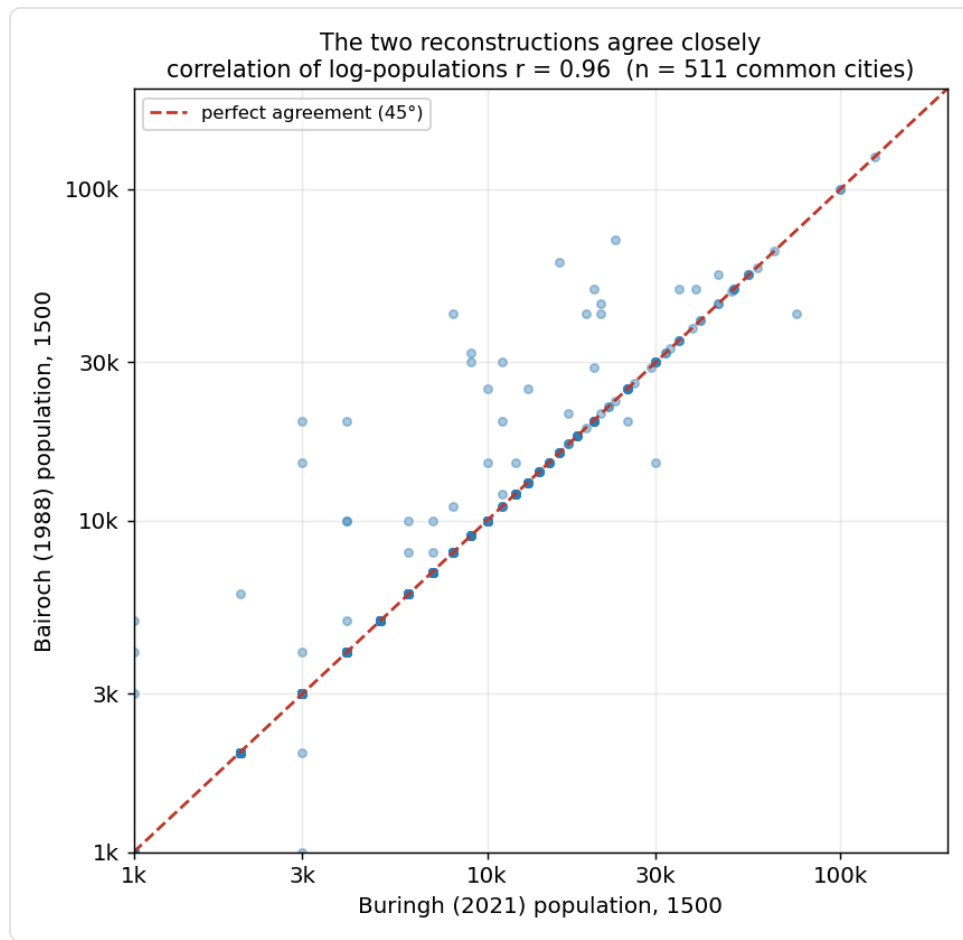


Figure 8. City populations in 1500: Bairoch (1988) versus Buringh (2021), for the 511 cities both cover. The log-populations correlate at $r = 0.96$; points hug the 45° line.

More importantly, the *conclusions* replicate. Figure 9 and Table 4 place the key quantities side by side. In both reconstructions, century-to-century growth is largely unpredictable, inherited size dominates the size distribution, water access carries a positive premium, and — decisively — the large raw "staple advantage" collapses toward zero once the causal comparison is made.

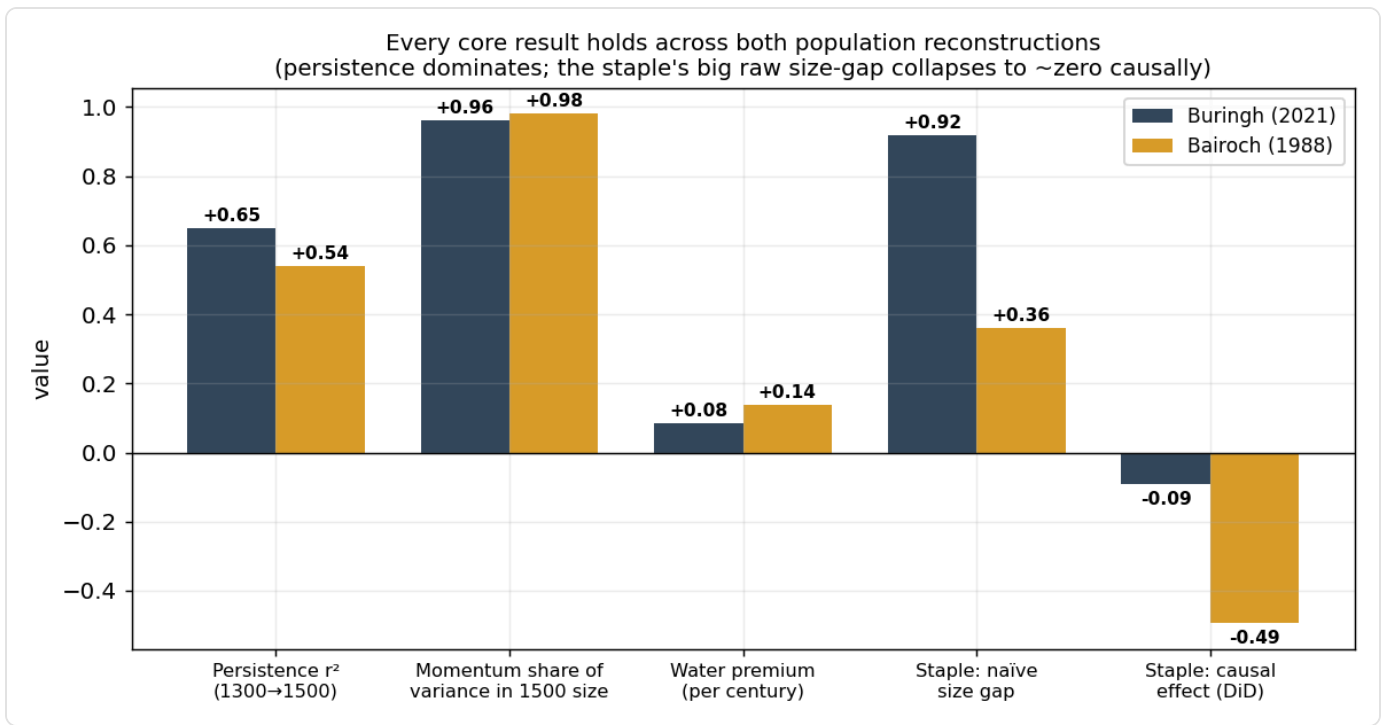


Figure 9. The same five quantities estimated on each population source. The pattern is identical: history (persistence, momentum) dominates; the staple's raw size gap is large but its causal effect is not.

	Result	Buringh (2021)	Bairoch (1988)
Growth vs. size, R^2 (1300→1500) — small = near-random		0.05	0.30
Persistence r^2 (1300→1500)		0.65	0.54
Momentum share of variance in 1500 size		96%	98%
Water-access growth premium / century		+9%	+15%
Staple right — raw size gap		+92%	+36%
Staple right — causal effect (DiD)		-9%	-49%

Table 4. Core findings across both sources. The two panels differ in coverage — Bairoch's sample skews toward larger, better-documented towns and is thinner before 1300, which is why its growth shows somewhat more mean-reversion and its causal staple estimate rests on only 7 treated towns (versus 53 in Buringh) — but the direction of every result is the same. The privileges' raw association with size is positive and large in both; where the data can support a well-powered causal test, it vanishes.

Robustness. The findings do not depend on which reconstruction one trusts. Across the two most widely used independent estimates of European city populations — Bairoch's foundational compilation and Buringh's georeferenced revision — history dominates

geography's direct role, water is the exogenous lever, and commercial privileges show no causal growth effect.

8. What this means for how we explain medieval cities

The story that a wise grant of market or staple rights conjured a great city reverses cause and effect. Rulers granted privileges to places that were *already* becoming important — usually because they sat on a river or coast, at a junction laid down in Roman or Carolingian times. The privilege was the *recognition* of urban success, drawn up after the fact, not its engine. This does not make institutions unimportant to medieval life; it means they are poor candidates for the *root cause* of differential city growth, because they are themselves consequences of the geography and history that did the causing.

The positive account is simpler and older than any charter: a city's fortune was set by **where** it sat in the landscape of trade — above all, whether goods could reach it by water — and by **how early** it had established itself, since size begat size across centuries. Policy operated at the margins of a system whose broad shape was already fixed.

9. Limitations

- **Rounded populations.** Pre-modern population figures are estimates, often rounded to round numbers. This inflates apparent randomness and caps every growth model's fit; our growth magnitudes should be read as orders of magnitude, not precise decimals.
- **Centennial resolution.** The population snapshots are a century apart, so our difference-in-differences uses century-wide "before" and "after" windows. The *null* result is robust across grant-cohorts and to the pre-trend check, but finer-dated wage or construction series would sharpen it.
- **Network centrality.** Graph-theoretic measures of position in the Viabundus network (betweenness, market access) did not out-predict a simple river/coast indicator within the Hanseatic footprint — a null worth stating plainly, and a caution against over-engineering the geography variable.

References

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Reproducibility: all figures, tables and estimates are generated by the scripts in `analysis/ ( panel.py · bairoch_panel.py → diagnostics.py → network.py → privileges.py → charter_did.py → causal_consolidated.py → synthesis.py → generative_model.py → bairoch_robustness.py → city_table.py → make_figures.py · figures2.py · figures3.py → build_paper.py )`.